

Notations:

prior distribution for Θ - f_{Θ}

conditional distribution for X given Θ - $f_{X|\Theta}$

posterior distribution for Θ given x - $f_{\Theta|x}(\cdot|x)$

Bayes rule:

$$\rightarrow f_{\Theta|x}(\theta|x) = \frac{f_{\Theta}(\theta) \cdot f_{X|\Theta}(x|\theta)}{\int_{\Theta'} f_{\Theta}(\theta') f_{X|\Theta}(x|\theta') d\theta'}$$

for continuous Θ

$$\rightarrow f_{\Theta|x}(\theta|x) = P(\Theta = \theta|x) = \frac{P(\Theta = \theta) \cdot f_{X|\Theta}(x|\theta)}{\sum_{\Theta'} P(\Theta = \theta') \cdot f_{X|\Theta}(x|\theta')}$$

for discrete Θ

Note: $f_{X|\Theta}(x|\theta) = P(X=x | \Theta=\theta)$, if conditional distribution of X given Θ is discrete

Estimators:

→ MAP estimator

$$\tilde{\Theta} = \underset{\Theta}{\operatorname{argmax}} f_{\Theta|x}(\theta|x)$$

→ Conditional expectation estimator

$$\tilde{\Theta} = E(\Theta | X=x)$$